PriceWise Pipeline — US Market

The pipeline in `pricewise_pipeline.ipynb` runs end to end on the US valid-sales file. It ends with a recommended price for each of 30 US pilot products: 7 increases, 9 decreases and 14 holds, worth about +$52k margin and −$42k revenue over 28 days. The frozen US baseline forecasts 28-day demand with 38.5% WAPE, ahead of every simple benchmark (best: last 7 days, 39.4%).

Pipeline at a glance

The input is `US_valid_sales_all_transactions.csv`: 2,717,497 US enterprise Standard Invoice lines (3.86M units, $2.14B revenue), Apr 2023 – Aug 2026. The US is modelled separately from Canada, with the same code and thresholds. Outputs are in `outputs/US/`.

Stage	Steps	What it produced
1 Data preparation	Clean & label → product eligibility → family diagnostics → pilot 30 → daily panel	331 of 4,612 products eligible; 30 pilot products; daily panel of 30 products × 1,246 days
1.5 Global Baseline v1	Simple benchmarks → learn global weights → time-based validation → freeze	`baseline_v1.json`: 38.5% WAPE vs 39.4% for the best benchmark
1.8 Baseline v2	Test segmentation within the US and CA vs US → compare weights → freeze	`baseline_v2.json`: no within-US split helped; own US weights beat pooled CA+US weights
2 Pricing	Business signals → objective weights → price adjustment → guardrails → final price	`s2_price_recommendations.csv`: 7 up, 9 down, 14 hold

The baseline answers "how many units will this product sell in the next 28 days at today's price?" Stage 2 then asks what happens to margin, revenue and volume if the price moves, and picks the best move within safe limits.

Stage 1 — Data preparation

Clean and label

No rows are deleted. Every line gets label columns, and later steps filter on them.

Label	Lines	Rule
Negative margin	409,332	margin < 0 (kept, flagged)
Bundled price	201,109	Unit price under $1, mostly $0.01 Services lines
Exact duplicate	111,753	Kept, per the data-prep note
Coverage gap	99,242	Line comes from the FY26Q3 file (Oct–Dec 2025)
Non-priceable type	78,494	Shipping, Other or missing product type
Price outlier	41,548	Outside 5× IQR of the product's own log unit price
Zero cost	11,872	cost ⩽ 0

Coverage gap. The FY26Q3 file (Oct–Dec 2025) has a median of 3 active customers a week, against 154 the quarter before and 194 the quarter after. Median weekly units fall about 70% (6,833 vs 20,000–25,000). This is missing data, not lost demand, and it's the same gap Canada has. Those days are set to unknown (not zero), are left out of rolling averages, and no forecast that touches them is scored.

The active customer base also grows steadily, from 67 a week in early FY24 to 204 by FY27. Volume per week stays flat, at about 20,000–25,000 units.

`invoice_revenue_country_currency` equals `revenue` in the US file, so `revenue` is USD.

Product eligibility (331 of 4,612)

A product-market is product × country. There are two lifecycle tiers, because laptop models sell for only about 12–15 months. Phones and Tablets, which only the US has, count as hardware.

Test	Run-rate (Options, Services)	Lifecycle (Notebook, Workstation, Desktop, Visuals, Servers, Phones, Tablets)
Active months	⩾ 24	⩾ 9

Test	Run-rate (Options, Services)	Lifecycle (Notebook, Workstation, Desktop, Visuals, Servers, Phones, Tablets)
First sale by	Aug 2024	Sep 2025
Total units	⩾ 500	⩾ 300
Sold in last 90 days	yes	yes
Bundled-price units	⩽ 10%	⩽ 10%
Distinct price points	⩾ 10	⩾ 10
Price CV	0.03–0.60	0.03–0.60

Main reasons products failed (first failed test): too few active months 2,317; launched too recently 963; no sale in 90 days 714; too few units 184; mostly bundled 72.

Family diagnostics

Family	Types	Products	Revenue ($M)	Margin %	Nov. index	YoY units	Eligible
Frame Road	Notebook	569	788.6	28.1%	1.12	−8%	35
Solar Peak	Notebook, Workstations	727	596.1	19.4%	0.92	−43%	30
Knight Thrive	Notebook	509	315.5	14.0%	0.84	+15%	33
Romeo Iris	Options, Services	1,627	190.6	49.6%	0.85	−12%	185
Titan Onset	Desktop	336	123.5	18.7%	0.57	−35%	38
Quest Lunar	Visuals	141	38.1	23.6%	0.95	+23%	9
Circuit Insight	Servers	118	10.7	36.5%	1.20	−29%	0
Harbor Momentum	Notebook, Servers, Services	409	2.2	30.5%	0.98	+80%	1
Signal Road	Options	89	0.3	−24.3%	1.11	+6%	0

Nov. index = November's share of units × 12 (1.0 = no seasonality). YoY compares Apr–Aug 2026 with Apr–Aug 2025.

US seasonality is weaker than Canada's: most families sit near or below 1.0. Signal Road again sells at a loss. Three tiny families (Stream Juliet, Lantern Scale, Sage Legend) are omitted.

Pilot 30 selection

The rule is the same as Canada's: each product type gets at least 2 slots (or all its eligible products if it has fewer), and the rest follow each type's share of eligible revenue. Within a type, the top sellers by units fill its slots.

Type	Eligible	Share of eligible revenue	Pilot slots
Notebook	68	50.2%	15
Workstations	30	19.6%	5
Services	98	12.4%	3
Options	87	9.2%	2
Desktop	38	7.4%	2
Visuals	9	1.2%	2
Servers	1	0.0%	1

Daily training panel

This is 30 products × 1,246 days (37,380 rows). The columns are the same as Canada's, with US federal holidays in place of Canadian ones. Gap days and days before a product's launch are blank, not zero.

Stage 1.5 — Global Baseline v1

The learned blend beats every simple benchmark, overall and in both tiers: 38.5% WAPE vs 39.4% for last-7-days. Unlike Canada, where last-28-days stayed ahead on accessories and services, the blend wins everywhere in the US.

How the baseline works

- **Target:** units sold in the next 28 days, forecast every Monday ("origin") from Apr 2024 onward.

- **Components:** six recent sales rates, each using only days before the origin and scaled to 28 days: last 7, 28, 91, 182 and 365 days (`ma7` … `ma365`), plus the same 28 days last year (`ly28`).
- **Baseline** = weighted sum of the six components. Weights are ⩾ 0 and add up to 1.
- **Fitting:** the weights minimise total absolute error pooled across the 30 US products, which is the same as minimising WAPE. This is solved exactly as a linear programme.
- **Missing history:** if a component has under 50% known days, it falls back to the next shorter window (ma365 → ma182 → ma91 → ma28; ly28 → ma91), so new laptops use the same weights.

Time-based validation

This is an expanding window: for each test quarter, the weights are learned only from forecasts whose 28-day target finished before that quarter began, then scored on the quarter. The test quarters are Jan, Apr and Jul 2025 and Jan, Apr and Jul 2026; Oct 2025 has no scorable rows because of the gap. That gives 1,526 scored product-weeks out of 1,929 usable rows.

Model	WAPE all	WAPE lifecycle	WAPE run-rate	MAE (units)	Bias
Global v1 (learned blend)	**38.5%**	**53.3%**	**23.3%**	191.8	−1.9%
Last 7 days	39.4%	54.8%	23.6%	196.2	−1.2%
Last 28 days	41.3%	57.1%	24.9%	205.5	−1.6%
Equal weights	41.3%	56.0%	26.1%	205.6	−1.1%
Last 91 days	45.2%	62.9%	27.0%	225.1	+0.9%
Last 182 days	47.0%	63.3%	30.3%	234.1	+1.5%
Last 365 days	50.4%	66.0%	34.4%	251.2	−0.4%
Same 28 days last year	57.0%	67.3%	46.3%	283.6	−5.9%

Lower is better. WAPE = total absolute error ÷ total actual units. MAE is larger than Canada's because US volumes are about 5× higher; WAPE is the fair comparison.

Learned weights

Weights learned in	Train rows	ma7	ma28	ma91	ma182	ma365	ly28
Frozen v1 (all data)	1,929	**0.52**	**0.25**	0.12	0.00	0.00	0.11
Jul 2026 fold	1,689	0.47	0.12	0.21	0.10	0.00	0.10

Weights learned in	Train rows	ma7	ma28	ma91	ma182	ma365	ly28
Apr 2026 fold	1,299	0.43	0.19	0.28	0.00	0.00	0.10
Jan 2026 fold	1,033	0.29	0.17	0.42	0.00	0.00	0.11
Jul 2025 fold	740	0.31	0.17	0.43	0.00	0.00	0.09
Apr 2025 fold	533	0.32	0.11	0.50	0.00	0.00	0.08
Jan 2025 fold	347	0.26	0.67	0.00	0.00	0.00	0.07

The most recent week carries the most weight (0.52), and that weight climbed from about 0.3 through 2025 to 0.47 by Jul 2026, as weight moved off the 91-day average. Year-long history adds nothing. Same-period-last-year keeps a steady ~0.10, a small seasonal correction. The average spread of each weight across folds is 0.09.

Stage 1.8 — Segmentation and Baseline v2

Country is the only segmentation that helps: the US does better with its own weights than with weights shared with Canada. No split within the US helped, so the US Baseline v2 is one profile with the v1 weights.

Within-US schemes

Each scheme learns its own weights per segment, using the same time-based folds as v1. A segment with fewer than 4 pilot products uses the global weights. To be adopted, a scheme had to cut WAPE by at least 2% (relative) vs v1, and keep its weights stable across folds (average spread ⩽ 0.20).

Scheme	Segments (with own weights)	WAPE	Change vs v1	Weight spread	Adopted
Product family	7 (4)	38.93%	+1.1%	0.12	No: worse
Lifecycle tier (lifecycle vs run-rate)	2 (2)	39.20%	+1.8%	0.11	No: worse
Product type	7 (2)	39.48%	+2.5%	0.11	No: worse
Demand pattern (smooth / erratic / lumpy)	3 (3)	39.52%	+2.6%	0.12	No: worse

The segments do learn different shapes. Frame Road laptops lean on the last week (ma7 0.66); Knight Thrive laptops lean on the 182-day average (0.69); accessories and services split between the last week and 91 days. With 30 products, though, these differences don't carry over to unseen quarters: every split is less accurate than the single global blend.

Country test: US weights vs pooled CA + US weights

The same folds were scored two ways: each country with its own weights, and both countries with one pooled set of weights. The pooled set is fitted so the larger US market can't swamp Canada.

Country	Own weights	Pooled CA + US weights
US	**38.52%**	39.59%
Canada	45.30%	45.31%

Own weights cut US WAPE by 2.7% (relative), which passes the 2% bar, and Canada is unaffected. So the country split is adopted: each market keeps its own baseline, as in the original data-prep plan. The weights show why: the US puts 0.52 on the last week, Canada only 0.34. Output: `outputs/s18_country_comparison.csv`.

Stage 2 — Final recommended price

Of the 30 US pilot products, 16 get a price move and 14 are held. Most holds are guardrails at work: 11 products have stopped selling, or sell below cost. Expected effect over the next 28 days: about +$52k margin and −$42k revenue (as of 2026-08-28, the last data date).

Pricing logic (same as Canada)

1. **Business signals** (per product):
   - Current price: the last 28 days' ASP.
   - Unit cost and margin %.
   - Baseline v2 units for the next 28 days.
   - Momentum and YoY units.
   - Price position vs the 365-day median.

    - Top-customer share and days since last sale.
    - **Price elasticity**:
        - A weekly log-log regression per product.
        - Shrunk toward its product type's pooled estimate. Servers, the one type with a single pilot product, is pooled with the other hardware products.
        - Bounded to [−4, −0.2].
2. **Objective weights** (placeholders, to agree with Bob): margin 0.40, revenue 0.25, volume 0.20, stability 0.15.
3. **Price adjustment:**
    - For each move from −15% to +15%: expected units = baseline × (1 + move)^elasticity.
    - Each objective is scored as % change vs holding the price.
    - The move with the highest weighted score wins.
4. **Guardrails**, applied in order:
    a. No sale in 28+ days → hold and review.
    b. Fewer than 20 units in 91 days → hold.
    c. **Price below unit cost → hold and review** (added for the US).
    d. Step cap ±10%, or ±3% when elasticity confidence is low.
    e. Trailing-365-day price band and cost + 5% floor; these can only shrink a move, never create one or exceed the cap.
    f. Round toward the current price.
    g. Moves under 1% after rounding → hold.

Data issues the guardrails caught

- `7D9EWDV7NH` **(Servers):** unit cost jumps from about $90 to $999 from Q1 2026 while price stays around $155, so the margin reads as −586%. This looks like a cost-data error (a decimal shift?), not a pricing problem.
- `36N5074JFA` **(Workstations):** sold 912 units in Aug 2026 at an average of $928, below its $974 unit cost. This is either a deliberate promotion or a cost issue; either way, a person should decide.
- **End-of-life:** 9 products haven't sold in 43–85 days, including five laptops, two workstations and two accessories.
- **Elasticity:** 10 of the 30 estimates hit the −4 floor and only 6 are high-confidence, so nearly every move is capped at ±3%.

Recommendations

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
03DF114WYO	Workstations	$1,857.39	$1,913.00	+3.0%	−1.84	step cap ±3% (low confidence)	+14,407
89UY8827CW	Notebook	$1,076.20	$1,108.00	+3.0%	−3.06	step cap ±3% (low confidence)	+13,282
95SS294NWP	Notebook	$1,069.15	$1,101.00	+3.0%	−4.00	step cap ±3% (low confidence)	+12,784
43HD010UKP	Workstations	$2,796.38	$2,880.00	+3.0%	−0.20	step cap ±3% (low confidence)	+5,561
04B5EGP6MO	Visuals	$85.16	$93.60	+9.9%	−0.20	step cap ±10%	+4,303
19Y55740YG	Notebook	$910.66	$937.00	+2.9%	−3.44	step cap ±3% (low confidence)	+1,437
06AS870MKS	Desktop	$1,068.89	$1,100.00	+2.9%	−2.04	step cap ±3% (low confidence)	+1,430
17HFYLR9WD	Notebook	$2,513.06	$2,438.00	−3.0%	−4.00	step cap ±3% (low confidence)	+6,288
5YO5U33136	Services	$103.04	$100.00	−2.9%	−3.14	step cap ±3% (low confidence)	+1,365
5IQ5M98979	Services	$207.29	$202.00	−2.6%	−4.00	step cap ±3% (low confidence)	+927
9TE3M17113	Services	$166.62	$162.00	−2.8%	−1.18	step cap ±3% (low confidence)	+553
68ZR1644BY	Notebook	$2,080.13	$2,018.00	−3.0%	−4.00	step cap ±3% (low confidence)	+407

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
78Z0DBV9SE	Visuals	$173.91	$157.00	−9.7%	−2.19	step cap ±10%	−836
41GEFQX2KE	Notebook	$1,135.97	$1,102.00	−3.0%	−3.74	step cap ±3% (low confidence)	−1,367
37NELHX9FG	Notebook	$1,892.08	$1,836.00	−3.0%	−4.00	step cap ±3% (low confidence)	−2,028
14XOTJR8EF	Notebook	$1,576.94	$1,530.00	−3.0%	−2.00	step cap ±3% (low confidence)	−6,605
36N5074JFA	Workstations	$929.19	hold	0%	−3.09	price below unit cost	0
7D9EWDV7NH	Servers	$143.57	hold	0%	−0.90	price below unit cost	0
17D4481BEM	Notebook	$1,027.58	hold	0%	−3.96	change under 1%	0
71SI047PCZ	Notebook	$1,052.83	hold	0%	−3.91	change under 1%	0
41YY0220YR	Desktop	$1,609.92	hold	0%	−1.94	none: objectives balance	0
76E1152PPM	Workstations	$3,799.00	hold	0%	−4.00	no sale in 85 days	0
27WH3132DN	Options	$131.62	hold	0%	−0.54	no sale in 80 days	0
00RN3110HH	Notebook	$2,433.62	hold	0%	−3.56	no sale in 79 days	0
04OA900WKL	Notebook	$2,659.00	hold	0%	−3.40	no sale in 79 days	0
30TF4746EQ	Workstations	$1,806.76	hold	0%	−2.23	no sale in 74 days	0
21WO7047EX	Notebook	$2,270.24	hold	0%	−3.51	no sale in 65 days	0

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
55C97203QD	Options	$279.39	hold	0%	−1.67	no sale in 65 days	0
99FL811XLB	Notebook	$2,340.15	hold	0%	−4.00	no sale in 52 days	0
16CHOUW4TD	Notebook	$1,849.95	hold	0%	−4.00	no sale in 43 days	0

Increases mostly go to laptops with thin margins (9–12%: `95SS294NWP`, `19Y55740YG`, `89UY8827CW`) and to products whose demand barely reacts to price (elasticity −0.2: `43HD010UKP`, `04B5EGP6MO`).

Decreases go to higher-margin products, from 19% for laptops to over 90% for services, where demand looks very price-sensitive. Several cuts still raise margin dollars, e.g. `17HFYLR9WD` at −3% gives +$6.3k, because an elasticity near −4 means many more units. These are the moves to test first, since they depend most on the elasticity estimate. The "why" text for each product is in the `rationale` column of the CSV.

US vs Canada

The US baseline is more accurate (38.5% vs 45.3% WAPE) and leans harder on the most recent week. Both markets share the same data gap, the same weak elasticity estimates and heavy end-of-life churn in the pilot.

	US	Canada
Transaction lines	2,717,497	498,263
Products / eligible	4,612 / 331	3,715 / 118
Median weekly active customers	66 → 207 (growing)	~25
Oct–Dec 2025 coverage gap	Yes (~3 customers/week)	Yes (~2 customers/week)

	US	Canada
Pilot 30 by type	15 Notebook, 5 Workstations, 3 Services, 2 each Options, Desktop, Visuals, 1 Servers	17 Notebook, 4 Options, 3 Services, 2 each Workstations, Desktop, Visuals
Baseline v1 WAPE	38.5%	45.3%
Best simple benchmark	Last 7 days, 39.4%	Last 28 days, 45.9%
Frozen weights (ma7 / ma28 / ma91 / ly28)	0.52 / 0.25 / 0.12 / 0.11	0.34 / 0.27 / 0.27 / 0.07
Segmentation adopted	Country only	Country only
Recommendations (up / down / hold)	7 / 9 / 14	10 / 10 / 10
Pilot products no longer selling	9	8
Below-cost products held	2	0
High-confidence elasticities	6 of 30	4 of 30
Expected 28-day margin / revenue change	+$52k / −$42k	+$13.3k / −$5.2k

The Canada details are in PriceWise Pipeline — What's Been Done.

Limitations and open decisions

Limitations

- **Elasticities are observational.** Within a week, price differences come from customer mix and volume discounts, which probably makes laptops look more price-sensitive than they are. In the US, 10 of 30 estimates hit the −4 floor and only 6 are high-confidence. Confirm them with controlled price tests, starting with the price cuts that are expected to raise margin.
- **Pilot churn.** 9 of the 30 pilot products haven't sold in 6–12 weeks, so a fixed pilot goes stale fast. A monthly refresh (like the PDF's dynamic top 30) would fix this.

- **Cost data.** At least one product (`7D9EWDV7NH`) has an implausible unit cost. A cost-data check should run before any price goes live.
- **Survivorship bias.** The pilot was picked using the full history (as a business pilot), so its validation slightly favours products that survived.
- **Small sample for segmentation.** With 30 products, no within-US split beat the global weights.
- **Missing signals.** No competitor prices (Dell, HP, Acer) and no inventory are in the data.
- **Negotiated prices.** B2B prices vary by customer, so the "current price" is an average, not a list price.

Decisions for the team

- [ ] Agree the objective weights with Bob (currently margin 0.40, revenue 0.25, volume 0.20, stability 0.15).
- [ ] Confirm the guardrails: ±10% step, ±3% at low confidence, cost + 5% floor, hold when below cost.
- [ ] Get `7D9EWDV7NH` 's cost data checked, and confirm whether `36N5074JFA` 's below-cost August sales were an intended promotion.
- [ ] Choose the pilot design: fixed vs refreshed monthly, and mixed vs hardware-only.
- [ ] Confirm the eligibility thresholds (24 months for run-rate products, 9 for lifecycle products).

Suggested slide outline

This is a 12-slide US deck. Each slide lists its title, then the message and where its content comes from in this doc. If the deck covers both markets, slides 11 and 12 carry the comparison.

1. **PriceWise US: from transactions to a recommended price**
   - Title slide.
   - The four stages, one line each (Pipeline at a glance).
2. **US data and what we cleaned**
   - 2.7M invoice lines, Apr 2023 – Aug 2026.
   - Nothing deleted, everything labelled (cleaning table).
3. **The Oct–Dec 2025 data gap**
   - About 3 active customers a week vs 150–200 around it.

- Treated as missing, not as zero sales.
- Visual: median weekly active customers by quarter (`s1_coverage_by_quarter.csv`).

4. **Which products can we price?**
   - 331 of 4,612 eligible.
   - Two lifecycle tiers; the main fail reasons.
5. **Family diagnostics**
   - Frame Road laptops carry the most revenue at 28% margin.
   - Solar Peak volume is down 43% YoY.
   - Romeo Iris services and accessories run at 50% margin.
6. **Pilot 30**
   - 15 laptops and 5 workstations, plus every other category (pilot table).
7. **The baseline model**
   - A weighted blend of six recent sales averages, with weights learned by minimising WAPE.
   - Visual: diagram of the components feeding the weighted sum.
8. **Does it work?**
   - 38.5% WAPE, beating every benchmark in both tiers.
   - Visual: bar chart of WAPE by model (`s15_validation_summary.csv`).
9. **Learned weights**
   - Half the weight on the last week.
   - Year-long history adds nothing (`s15_weights_by_fold.csv`).
10. **From forecast to price**
    - Signals, then objective weights, then the best move, then guardrails.
    - Visual: flow diagram (Pricing logic).
11. **US vs Canada**
    - The comparison table.
    - Separate weights per country beat shared weights (country test).
12. **Recommendations and next steps**
    - 7 up, 9 down, 14 hold; +$52k margin and −$42k revenue over 28 days.
    - Data issues flagged; decisions for Bob.

Files and how to rerun

Run:

1. Put both CSVs in your Downloads folder, with the US file unzipped.
2. Leave `DATA_FILES` in the Config cell pointing at `Path.home() / "Downloads" / ...`.
3. Run all cells top to bottom: about 2 minutes for both markets, or set `COUNTRIES = ["US"]` for the US only.

It needs `pandas>=2.2 numpy scipy statsmodels holidays pyarrow`; use `%pip install ...` in the first cell, then restart the kernel. Every threshold in this doc lives in the Config cell.

Output file (`outputs/US/`)	Stage	What's in it
`s1_coverage_by_quarter.csv`	1	Active customers per source quarter; gap flag
`s1_product_eligibility.csv`	1	Every product, each test pass/fail, first failed rule
`s1_family_diagnostics.csv`	1	One row per product family
`s1_pilot_30.csv`	1	The 30 pilot products and their attributes
`s1_daily_panel.parquet`	1	Product × day training panel
`s15_validation_summary.csv`	1.5	WAPE / MAE / RMSE / bias per model, overall and per tier
`s15_weights_by_fold.csv`	1.5	Weights learned in each test quarter
`baseline_v1.json`	1.5	Frozen US Global Baseline v1
`s18_scheme_comparison.csv`	1.8	Within-US segmentation results
`baseline_v2.json`	1.8	Frozen US Baseline v2
`s2_business_signals.csv`	2	All signals, including elasticity details
`s2_objective_grid.csv`	2	Score of every candidate price move
`s2_price_recommendations.csv`	2	Final prices, guardrails, expected impact, rationale